Build & Evaluate a Linear Regression Model

House Price Predictor · California Housing Dataset

1. Executive Summary

This report documents the complete ML workflow for predicting California median house values using Linear Regression. Data from the 1990 U.S. Census (20,640 records, 8 features) was explored, preprocessed with StandardScaler, and trained on an 80/20 train-test split. The final model achieved R^2 = **0.6634**, explaining **66.3%** of the variance in house prices, with a Mean Absolute Error of **$46,070**.

Metric	Value	Meaning
MAE	0.4607	Average prediction error ≈ $46,070
RMSE	0.5571	Penalizes large errors more than MAE
R^2 Score	**0.6634**	Model explains 66.3% of variance in house prices

2. Dataset Overview

The **California Housing Dataset** is sourced from the 1990 U.S. Census and built into scikit-learn. It has **20,640 records**, **8 numerical features**, and zero missing values. The target variable is the median house value (in $100,000 units) for each census block group.

Feature	Description	Unit
MedInc	Median household income in block group	$10,000
HouseAge	Median house age in block group	Years
AveRooms	Average number of rooms per household	Count
AveBedrms	Average bedrooms per household	Count
Population	Block group population	People
AveOccup	Average household occupancy	Ratio
Latitude	Block group latitude	°N
Longitude	Block group longitude	°E
MedHouseVal	**TARGET: Median house value**	$100,000

3. Exploratory Data Analysis

3.1 Correlation Heatmap

MedInc has the strongest positive correlation with house value. **Latitude** is negatively correlated — northern California generally has lower prices. AveRooms and AveBedrms show mild multicollinearity.

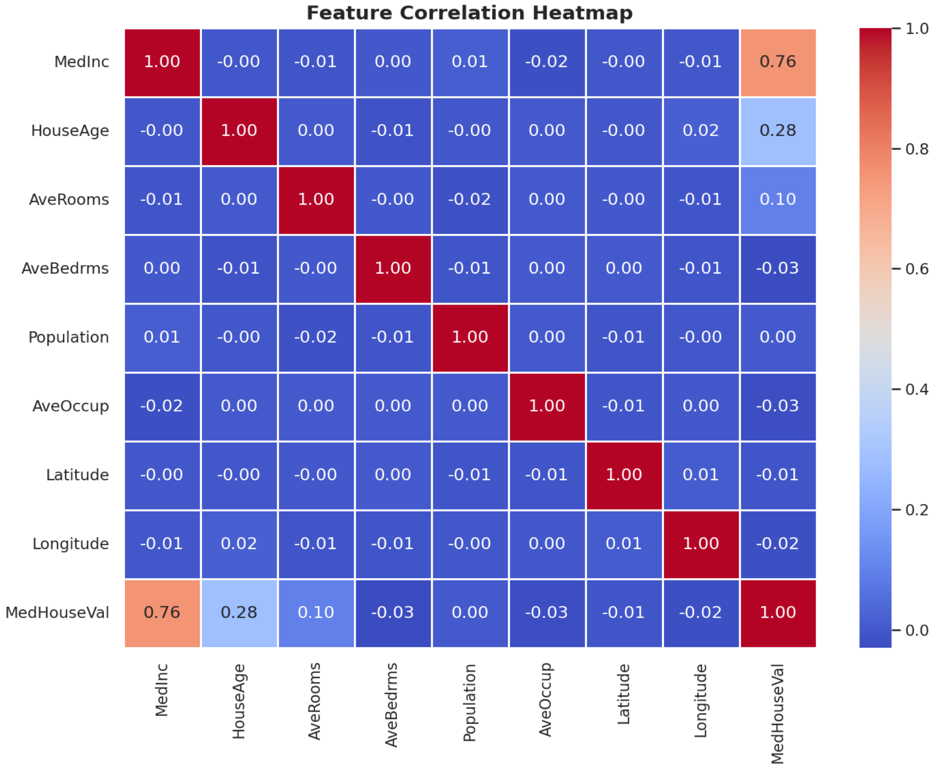

Figure 1 — Feature Correlation Heatmap

3.2 Target Variable Distribution

The distribution is right-skewed with a hard cap at $500,000 — a census artifact. Most properties fall in the $100k–$300k range.

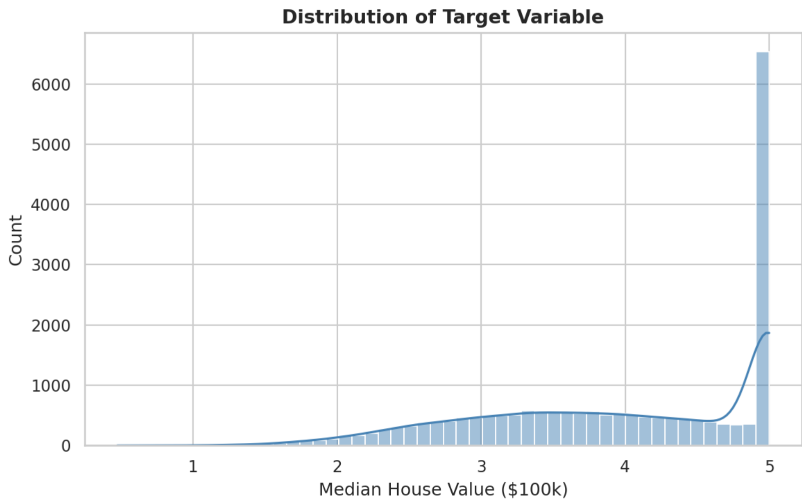

Figure 2 — Distribution of Target Variable (MedHouseVal)

3.3 Feature Distributions

Most features are right-skewed. Population and AveOccup have extreme outliers. Log-transforming these could improve model fit in future iterations.

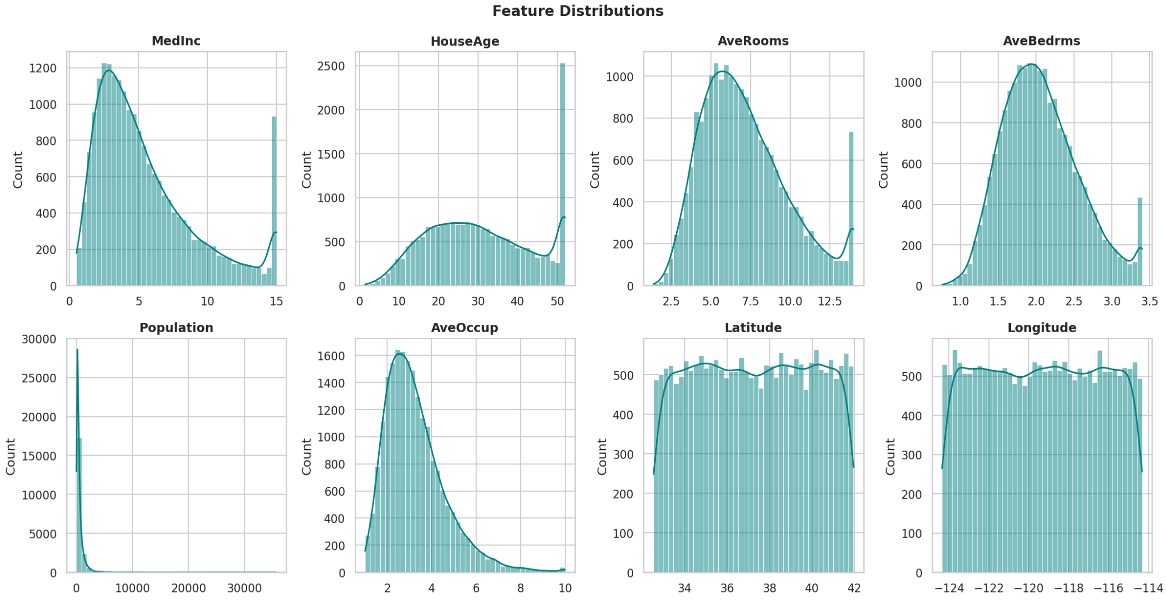

Figure 3 — All Feature Distributions

4. Methodology

Train / Test Split: 80/20 split with random_state=42 → 16,512 train / 4,128 test samples.

Feature Scaling: StandardScaler fitted on training data only (prevents leakage), then applied to test set.

Model: sklearn LinearRegression — Ordinary Least Squares with no regularization.

Evaluation: MAE, RMSE, and R^2 computed on the held-out test set.

5. Results

5.1 Actual vs. Predicted

Points near the red diagonal indicate accurate predictions. The fan-shape at high values reflects difficulty predicting expensive properties, partly due to the $500k ceiling.

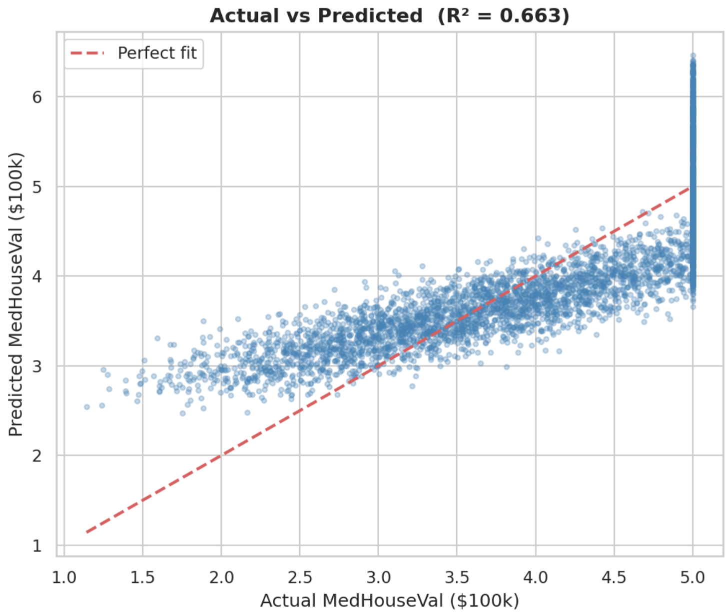

Figure 4 — Actual vs. Predicted Values (Test Set)

5.2 Residual Analysis

Residuals are approximately normally distributed with mean ≈ 0. Slight patterns at high fitted values suggest a non-linear component the model misses.

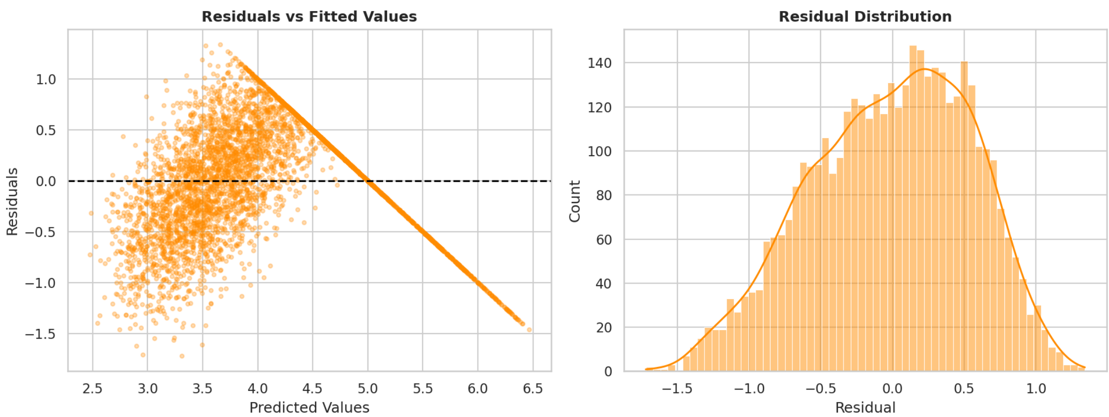

Figure 5 — Residuals vs. Fitted and Residual Distribution

5.3 Feature Importance (Coefficients)

Standardized coefficients reveal relative feature importance. **MedInc** (0.73) dominates all others. Geographic and structural features contribute much less, suggesting income alone drives most of the explainable variance.

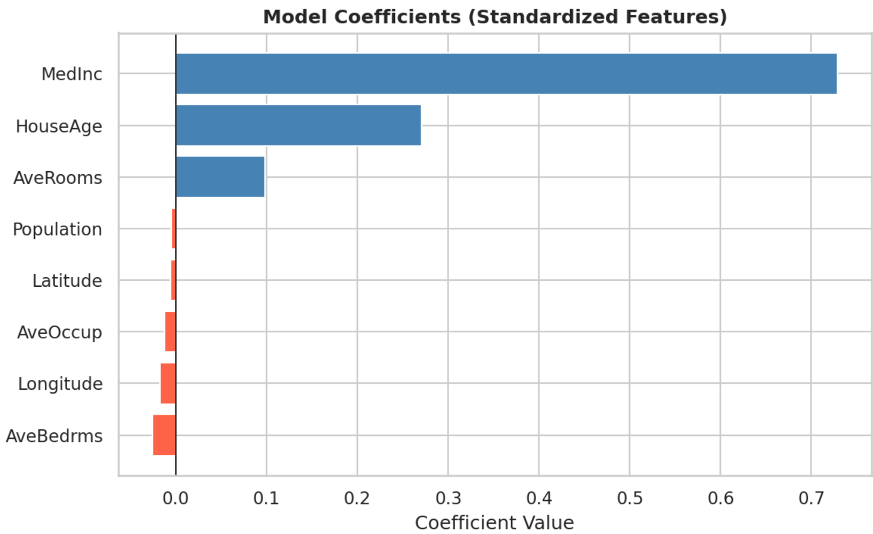

Figure 6 — Standardized Model Coefficients

Feature	Coefficient	Interpretation
MedInc	+0.7286	1 SD rise in income raises predicted value by 0.73 units (~$73k)
HouseAge	+0.2708	Older blocks show modest positive effect
AveRooms	+0.0985	More rooms per household increases value slightly
AveBedrms	-0.0257	High bedroom-to-room ratio mildly lowers value
Longitude	-0.0178	Eastern (inland) regions have lower prices
AveOccup	-0.0127	Higher occupancy (crowding) slightly reduces price
Latitude	-0.0060	Inland northern areas trend lower in price
Population	-0.0054	Near-zero — block population barely affects price

6. Improvement Ideas

1. Feature Engineering — Log-transform skewed features (Population, AveOccup). Create interaction terms (MedInc × HouseAge). Bin lat/long into geographic zones.

2. Polynomial Regression — Use PolynomialFeatures to capture non-linear relationships. Expected R^2 gain of 5–10% without changing the base estimator.

3. Ridge / Lasso Regression — Regularize to handle multicollinearity between AveRooms and AveBedrms. Lasso also performs implicit feature selection.

4. Gradient Boosting (XGBoost / LightGBM) — Tree-based models naturally handle non-linearities. Typically achieve $R^2 > 0.85$ on this dataset.

5. Cross-Validation — Replace single split with 5-fold CV for more reliable generalization estimates.

6. Outlier Handling — Cap extreme AveRooms (>20) and AveOccup (>10) values before training.

7. Conclusion

The Linear Regression baseline achieved R^2 = **0.6634** (MAE = $46,070, RMSE = $55,710). **MedInc** is the dominant predictor. Residual patterns suggest non-linear structure remains in the data. Future work should explore feature engineering and gradient boosting to significantly improve predictions beyond this linear baseline.